

Dr Loizos Pelecanos

MEng (Hons) ACGI PhD DIC FHEA MIMarEST

Department of Architecture & Civil Engineering, University of Bath

Civil Engineer & Academic Researcher

Marine & offshore infrastructure · computational geotechnics · structural dynamics · earthquake engineering · sensing · optimisation · AI/ML



35+	90+	5	59
journal papers published / accepted	research outputs	completed PhDs	MEng/MSc dissertations

RESEARCH PROFILE

Mechanics-led research across marine and offshore infrastructure, geotechnics and structural dynamics. The programme combines nonlinear computational modelling with field monitoring and experimental evidence, optimisation and data-driven methods, with applications to offshore wind, waterfront and buried infrastructure, foundations, dams, buildings, bridges and critical infrastructure.

CORE RESEARCH THEMES - Marine & offshore infrastructure and hydrodynamic loading - Computational geotechnics and soil-structure interaction - Earthquake engineering and structural dynamics - Structural health monitoring and distributed fibre-optic sensing - Sustainable foundation engineering and optimisation - Inverse analysis, AI/ML and model updating	ACADEMIC & PROFESSIONAL SERVICE - Editorial Board: Computers & Geotechnics; Scientific Reports - Invited Editor: special issue, Underground Space - Cyprus National Representative: CEN/TC 250/SC10/WG04 - UK BDS Representative: ICOLD Technical Committee A - Member, Institute of Marine Engineering, Science & Technology (MIMarEST) - Fellow of the Higher Education Academy (FHEA)
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ACADEMIC TRAJECTORY

2016	Appointed Lecturer in Civil Engineering - Department of Architecture & Civil Engineering, University of Bath
2013-2016	Postdoctoral Research Associate - Centre for Smart Infrastructure and Construction, University of Cambridge
2013	PhD in Computational Earthquake Engineering - Imperial College London
2009	MEng (Hons) in Civil Engineering - Imperial College London

RESEARCH LEADERSHIP

Five completed PhD students and 59 completed MEng/MSc dissertations. More than £265k in research and PhD-studentship funding, including Royal Society and EPSRC/UKRI support spanning resilient infrastructure, offshore energy, monitoring and lower-carbon foundation engineering.

Selected research evidence

A concise cross-section of published and field-based work across the research programme.

Marine / offshore wind	Williams, S., Pelecanos, L. & Darby, A. (2024). A practical two parameter model of pile-soil gapping for prediction of monopile offshore wind turbine dynamics. <i>Géotechnique</i> 74(3), 265-280.
Marine / long-term cyclic response	Williams, S.A., Pelecanos, L. & Darby, A.P. (2022). A Winkler model of monopile ratcheting under long-term dynamic cyclic loading. <i>Ocean Engineering</i> 266, 112625.
Hydrodynamics / waterfront structures	Pelecanos, L., Kontoe, S. & Zdravkovic, L. (2013). Numerical modelling of hydrodynamic pressures on dams. <i>Computers and Geotechnics</i> 53, 68-82.
Sensing / inverse analysis	Yang, Y., Shin, G.-B., Pelecanos, L., Wang, C.C., Luo, L. & Soga, K. (2026). Inferring pile capacity using distributed strain sensing with PDE-constrained optimization. <i>Computers and Geotechnics</i> 196, 108131.
Sensing / deep foundations	Shin, G.-B., Soga, K., Yang, Y., Pelecanos, L., Mevoli, F.A. & Yeskoo, A. (2026). Distributed Fiber Optic Sensing for Monitoring Piles under Eccentric Load. <i>ASCE Journal of Geotechnical and Geoenvironmental Engineering</i> 152(9), 04026089.
Earthquake engineering	Pelecanos, L., Kontoe, S. & Zdravkovic, L. (2015). A case study on the seismic performance of earth dams. <i>Géotechnique</i> 65(11), 923-935.
Sustainable foundations	Abushama, K., Hawkins, W., Pelecanos, L. & Ibell, T. (2024). Embodied carbon optimisation of concrete pile foundations and comparison of different pile geometries. <i>Engineering Structures</i> 310, 118109.

FIELD SENSING & ENGINEERING TRANSLATION

Distributed fibre-optic and SHM work has included Farringdon O-cell pile testing, London Bridge Station masonry arches, the U.S. Embassy London thermal pile, CERN TT10 tunnel monitoring, V&A; Museum tension piles, geothermal energy piles and steel structural members. The programme has progressed from measurement to mechanics-based interpretation, inverse load-transfer inference and capacity estimation.

TEXTBOOK

Kechavarzi, C., Soga, K., De Battista, N., Pelecanos, L., Elshafie, M.Z.E.B. & Mair, R.J. (2016). *Distributed Fibre Optic Strain Sensing for Monitoring Civil Infrastructure: A Practical Guide*. ICE Publishing / Thomas Telford.

CURRENT RESEARCH DIRECTIONS

Marine infrastructure Wave-soil-structure interaction, quay walls, coastal assets and multi-hazard resilience.	Monitoring-informed models SHM, modal identification, FE correlation, inverse analysis and digital models.
Offshore energy systems Long-term foundation response, wave-wind loading and soil-foundation-structure interaction.	AI-enabled mechanics Surrogates, physics-informed learning, optimisation and data assimilation for engineering decisions.

SELECTED FUNDED PROGRAMMES

Royal Society international exchanges on underground-infrastructure resilience, combined wind-wave energy systems and long-span bridge-cable performance; EPSRC/UKRI impact-acceleration work on lower-carbon foundations using Thermal Integrity Profiling.

ONLINE

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